

Small Defect Detection: Dataset Standardisation, Benchmarking and Visual-Language Annotation

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Why Automated Defect Inspection Matters

- Manual inspection is slow, inconsistent, and error-prone
- Small defects are easily overlooked by human inspectors
- Defects appear across diverse surfaces and product types
- Missing a defect impacts quality, safety, and manufacturing costs

The challenge: detecting defects that occupy as few as 10–20 pixels in a high-resolution image

The Technical Challenge

- Small defects occupy very few pixels — often <1% of image area
- Image downsampling during CNN processing destroys fine spatial detail
- Industrial surfaces contain noise and repetitive background textures
- Severe class and size imbalance across datasets
- Different datasets use incompatible annotation formats

Key Insight

A 4px scratch in a 200×200 image becomes sub-pixel after just 4 stride-2 CNN layers

Stride-2 layers:

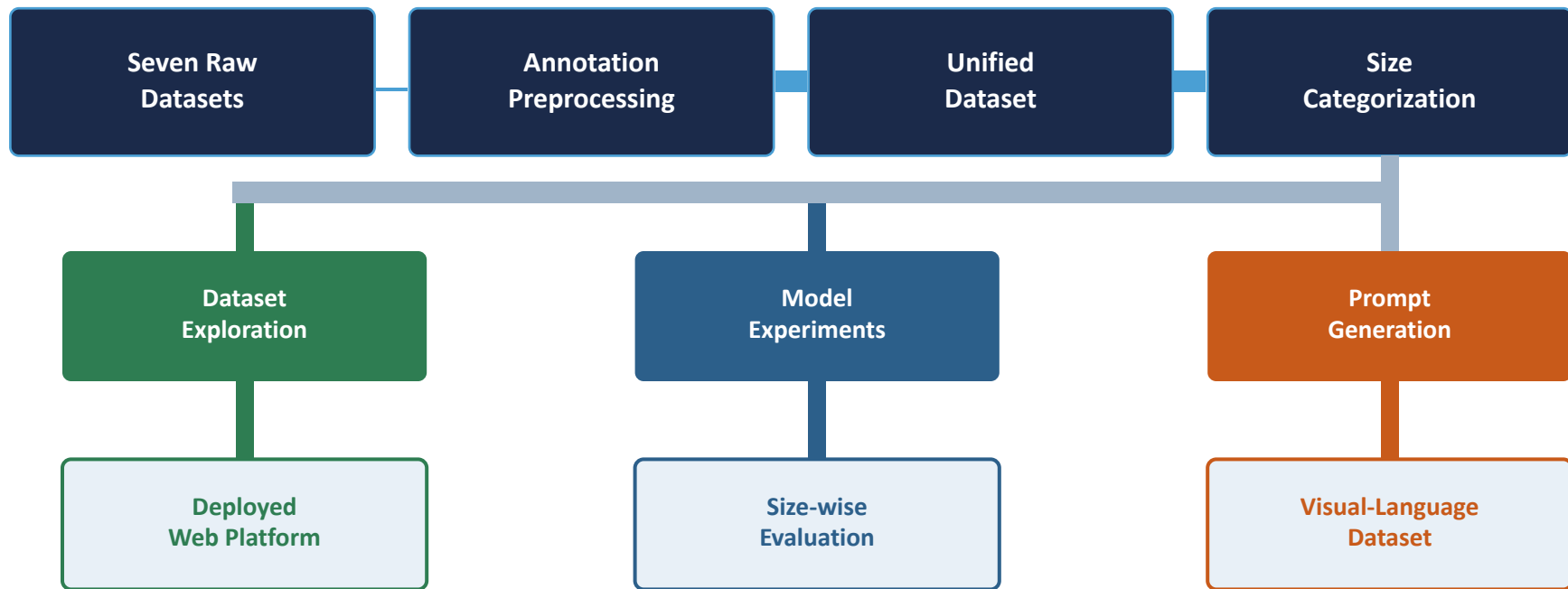
200→100→50→25→12 px

A 4px defect vanishes

Project Objectives

- 1 Combine multiple industrial-defect datasets from top research venues
- 2 Standardize images, masks, and annotations into a unified format
- 3 Categorize defects by relative size: small / medium / large
- 4 Build tools for dataset exploration and querying
- 5 Compare detection and segmentation model architectures
- 6 Generate natural-language descriptions for every defect instance
- 7 Deploy an interactive dataset exploration platform

End-to-End Pipeline



Each branch processes all seven datasets consistently through a single unified codebase

Seven Industrial Datasets

Dataset	Domain	Images	Source
DAGM	Textured surfaces	2,100	Workshop
GC10-DET	Steel strips	2,292	Journal
KolektorSDD2	Electrical commutators	356	Journal
MPDD	Metal parts	282	Journal
MTD (Magnetic Tile)	Magnetic tiles	388	GitHub
Severstal	Steel sheets	7,095	Kaggle
VisA	Mixed industrial	157	ECCV 2022
TOTAL	Seven domains	12,670	

Annotation Standardization Pipeline

Original Formats

- PASCAL VOC XML bounding boxes
- Binary PNG masks
- Weak elliptical masks
- RLE run-length encoded masks
- Dataset-specific category labels

Unified
Pipeline

Standardized Outputs

- YOLO .txt labels (cx, cy, w, h normalized)
- COCO JSON annotations
- Binary segmentation masks
- Size manifests (CSV)
- Unified class mappings

Size-Based Defect Classification

$$\text{Relative Defect Area} = \text{Bounding-box Area} \div \text{Image Area}$$

Category	Threshold	Images	% of Dataset
Small	< 1% of image area	3,035	24%
Medium	1% – 5% of image area	5,741	45%
Large	≥ 5% of image area	3,894	31%
Total	—	12,670	100%

Small defects (24% of data) are the primary research focus — hardest to detect, highest industrial risk

Interactive Dataset Exploration Platform

- Seven dataset cards with statistics
- Size-distribution visualisations
- Individual dataset pages
- Image browsing with bounding-box overlays

- Search and filter by class and size
- Downloadable CSV summaries
- Hugging Face dataset storage backend
- Deployed on Vercel (Next.js + TypeScript)

Fair Model Comparison Protocol

Experimental Setup

- Fixed dataset: 12,670 unified images
- Split: 70% train / 15% val / 15% test
- Seed 42 for full reproducibility
- Stratified by dataset and size category
- Same val/test subsets across all models
- Weights & Biases experiment tracking

Primary Metric

Small-Defect Recall

*Of all small defects in the test set,
what fraction does the model find?*

Also tracked: mAP@50 · mAP@50-95
Precision · Recall · Dice (segmentation)

Multi-Architecture Experiment Suite

Model	Type	Task	Key Feature
YOLOv8n	CNN detector	Detection	Baseline
YOLOv8n-P2/PAN	CNN detector	Detection	P2 head for small objects
YOLOv7	CNN detector	Detection	Architecture comparison
YOLOv10n	CNN detector	Detection	End-to-end, NMS-free
YOLOE11s	CNN detector	Detection + Seg	Prompt-aware
D-FINE-S	Transformer	Detection	Transformer comparison
SegFormer-B0	Transformer	Segmentation	Binary segmentation

All models evaluated on the same fixed test split for fair comparison

Natural Language Annotations for Every Defect

What Was Generated

- 27,230 prompt entries across all instances
- Progressive detail: p1 → p5 → p9
- Attributes: type, size, shape, count,
- location, severity, surface, domain

Example: Scratch on Steel Strip

p1

"A defect on a steel surface."

p5

"A small elongated scratch in the upper-left region of a steel strip surface."

p9

"A minor linear scratch defect at ~0.4% of image area, in the upper-left quadrant, oriented diagonally with sharp edges and low severity."

What Was Built

Key Contributions

- Unified 7-dataset industrial defect benchmark
- 12,670 standardised images
- 3,035 small-defect images with full annotations
- YOLO + COCO + segmentation formats
- Size-wise analysis and evaluation framework
- Interactive deployed web platform
- 27,230 visual-language prompt annotations
- Reproducible pipeline with fixed eval protocol

Next Steps & Conclusion

Future Work

- Complete all model experiments under fixed compute budget
- Publish consolidated benchmark comparison table
- Cross-dataset generalisation testing
- Tiled inference for high-res small defect detection
- Train language-guided segmentation models
- Add model predictions to deployed platform

Conclusion

Seven incompatible datasets converted into one standardised benchmark.

A reproducible framework established for small-defect detection evaluation.

Extended with an interactive platform and 27,230 visual-language annotations — providing an end-to-end foundation for future small industrial defect research.